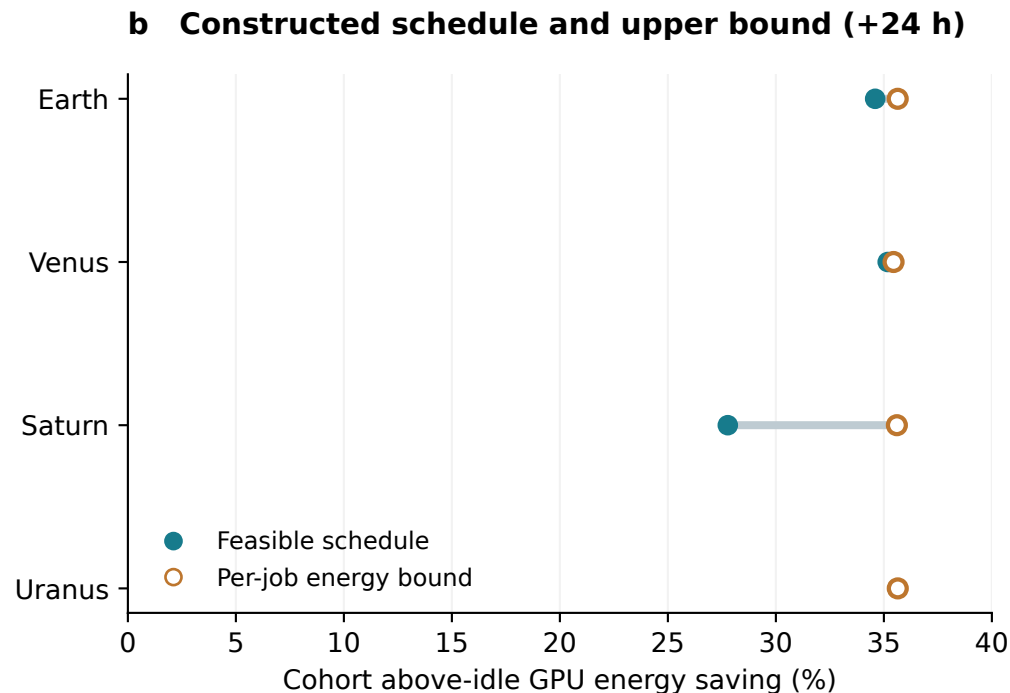
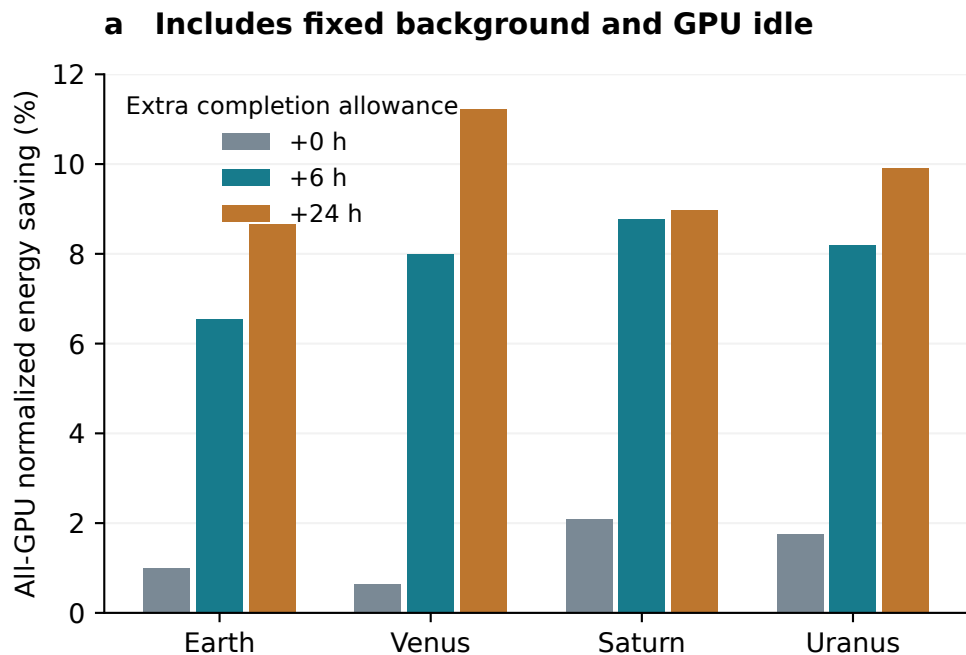


Service constraints shape the usable energy opportunity

Four production cohorts | non-preemptive fixed-size GPU gangs | retrospective completion benchmarks



Conditional GPU-only model: one common DVFS curve, assumed GPU idle ratio 0.10. No whole-node or grid benefit is measured.

Cohort: completed jobs submitted and finished during 6–13 April 2020; 192 h domain retains all other allocations as background.

Panel b brackets achievable and upper-bounded opportunity in this model; it is not a confidence interval or proof of global optimality.